COMP3311 Database Systems
Final Exam 21T1 (Sample Only)

The University of New South Wales

COMP3311 21T1 Database Systems
Final Exam (SAMPLE ONLY)

Friday 30 April 2021

1. Time Allowed: 3 hours
2. To be completed between:
   1:30pm to 6:00pm Fri 30 April 2021 (AEST)
3. Total number of questions: 11
4. Total marks available: 60
5. Questions are not of equal value.
6. Marks are shown on each question.
7. **Carefully read the notes below before you start the exam.**

By submitting the exam answers via GIVE, you declare that **all of the work submitted for this exam is your own work, completed without assistance from anyone else. You also declare that you will not copy or share the questions to anywhere outside CSE except onto your home computer, and you will promptly remove them after you submit the exam.**
Warning: We will be running plagiarism-checking on your submissions. Plagiarism, especially in exams, will be seriously prosecuted as Student Misconduct. Please refer to the Student Conduct web site for details.

1. General Instructions:

- Answer **all** questions.
- Questions are not worth equal marks.
- Questions may be answered in any order.
- You may create additional views to help formulating your queries, if needed, but you are not allowed to create any tables.
- During 1:30pm to 6:00pm Fri 30 April 2021 (AEST), you must **not**:
  - access any of your own files
  - access any web pages except the standard documentation in this course.
- During the 24-hr period from 1:30pm Fri 30 April 2021 to 1:30pm Sat 1 May 2021 (AEST), you must **not**:
  - communicate with other students in *any* way regarding this examination
- Your answers must be submitted using `give` or via WebCMS
- You can submit questions multiple times; only the last submission will be marked.
- Do not leave submissions to the last minute

Please fill in your answers in the supplied templates: `ans.sql` for the SQL queries and `ans.txt` for the written questions. The comments in the template files indicate where you can fill in the answers. If there are extra files (e.g., drawings) needed to be submitted, the corresponding questions will have instructions specified. SQL queries will be auto-marked by using PostgreSQL installed on `grieg`, on a database with the same schema with data possibly modified. You can only receive marks for correctly-working queries that loads with no errors. To enable auto-marking, your queries should be formulated as SQL views, using the view names and attribute names provided in `ans.sql` .

2. Submission:

You can submit your exam solution either using:

i) WebCMS: Login to Course Web Site > Exam > Final Exam > Final Exam Paper > Make Submission > upload required files > [Submit]

Or:

ii) The give command: `give cs3311 exam ans.sql ans.txt ans.pdf`

Required Files: `ans.sql ans.txt ans.pdf`

Deadline: Friday 30 April 2021 at 6:00pm (AEST)

No late submissions will be accepted.

3. Downloads:

Downloads: exam.tgz or exam.zip

Both `.tgz` and `.zip` files contain the same material.
Each archive contains the PostgreSQL beer database `exam.sql` (consists of the schema and the data), plus the answer template files called `ans.sql` and `ans.txt`

4. How to Start:

- read these notes carefully and completely
- download one of the archive files above
- unpack the downloaded file; exam.sql includes the schema and the data
- the schema in exam.sql is illustrated in the ER diagram that you should be familiar with from the prac exercises
- read `ans.sql` and `ans.txt`; and identify where you can fill in your answer for each question
- attempt the questions and fill in the answers in `ans.sql` and `ans.txt` accordingly
- you are allowed to create additional views to help to formulate your queries if needed, but you are not allowed to create any extra tables
- login to `grieg`, and test your `ans.sql`
- submit all the Required Files via WebCMS3 (or give) as described above

A basic test script for the SQL queries is available. Make sure you have started the PostgreSQL server on `grieg`, go insider the folder that contains your ans.sql and run the following command:
~cs3311/sample/autotest

End of Notes

Question 1 (5 marks)

Given the provided exam.sql, write an SQL query to find the brewers (output in ascending order) whose beers John likes.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 2 (5 marks)

Given the provided exam.sql, write an SQL query to find how many beers each brewer makes. The output is ordered by brewer in ascending order.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 3 (5 marks)

Given the provided exam.sql, write an SQL query to determine beers sold at bars where John drinks. The output is ordered by beer in ascending order.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 4 (5 marks)

Given the provided exam.sql, write an SQL query to determine what the most expensive beer is. In case of more than one most expensive beers, the output is ordered by beer in ascending order.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 5 (5 marks)

Given the provided exam.sql, write an SQL query to find the average price of common beers ("common" = served in more than two hotels). The output is ordered by beer in ascending order.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 6 (5 marks)

Given the provided exam.sql, write an SQL query to find name of cheapest beer at each bar. The output is ordered by bar and then by beer in ascending order.

Instructions:

- Your answer will be expressed as a view with its name and arguments already defined in `ans.sql`

Question 7 (4 marks)

Consider the following (slightly unusual) definition for a table of enrolment information in a student information system:

```
create table Enrolments (
        student_id  integer,
        course_code char(8),
        semester    char(4),
        prac_mark   integer,
        exam_mark   integer,
        final_mark  integer,
        grade       char(1),
        primary key (student_id, course_code, semester)
);
```

The table currently has no constraints (apart from the primary key) to ensure that the attributes have sensible values. Add constraints to ensure that each of the following conditions is satisfied:

a. student IDs are 7-digit numbers in the range 2000000 to 4999999 inclusive

b. course codes are like UNSW course codes (4 upper-case letters followed by 4 digits)

c. semesters are like UNSW semester codes (*YYsN* e.g. `'13s1'`, `'00s2'`, `'05x1'`, `'07x2'`)

d. the prac mark must be in the range 0 to 50 inclusive

Instructions:

- Please modify the dummy placeholder REPLACE ME of the corresponding answer in `ans.txt`

Question 8 (7 marks)

Consider a relation R(A,B,C,D). For each of the following sets of functional dependencies, assuming that those are the only dependencies that hold for R, list all of the candidate keys (separated by commas) for R.

a. C -> D, C -> A, B -> C

b. B -> C, D -> A

c. ABC -> D, D -> A

Instructions:

- Please modify the dummy placeholder REPLACE ME of the corresponding answer in `ans.txt`

Question 9 (7 marks)

Consider a relation R(A,B,C,D). For each of the following sets of functional dependencies, assuming that those are the only dependencies that hold for R, if R is not already in BCNF, decompose it into a set of BCNF relations (separated by commas). If it is already in BCNF, just write ABCD as the final relation (i.e., no need for any BCNF decomposition).

a. A -> BCD

b. ABC -> D, D -> A

Instructions:

- Please modify the dummy placeholder REPLACE ME of the corresponding answer in `ans.txt`

Question 10 (6 marks)

For each of the following schedules, determine if it is serializable.

a.
```
T1:                 R(X) W(X) W(Z)            R(Y) W(Y)
T2: R(Y) W(Y) R(Y)                  W(Y) R(X)            W(X) R(V) W(V)
```

b.
```
T1: R(X) R(Y) W(X)            W(X)
T2:                 R(Y)                R(Y)
T3:                      W(Y)
```

Instructions:

- Please modify the dummy placeholder REPLACE ME of the corresponding answer in `ans.txt`

Question 11 (5 marks)

Draw an ER diagram for the following application from the manufacturing industry:

- Each supplier has a unique name.
- More than one supplier can be located in the same city.
- Each part has a unique part number.
- Each part has a colour.
- A supplier can supply more than one part.
- A part can be supplied by more than one supplier.
- A supplier can supply a fixed quantity of each part.

Instructions:

- Save your drawing as a file called `ans.pdf` in PDF format.

End of Exam